

Weather Data Analysis and Rainfall Prediction

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Introduction

Weather conditions play a crucial role in many real-world decision-making scenarios, including agriculture, transportation, and daily planning. Accurate rainfall prediction can help mitigate potential risks and improve resource allocation. In this project, we aim to develop a rainfall prediction model based on historical weather observations.

Using a structured weather dataset containing daily meteorological measurements, we first perform data cleaning and exploratory data analysis to understand the distribution, patterns, and relationships among weather variables. We then engineer additional features and apply multiple machine learning models to predict whether it will rain on the following day. Through systematic analysis and modeling, this project seeks to identify key weather factors associated with rainfall and evaluate the predictive performance of different classification models.

Data Cleaning and Preprocessing

We first loaded the dataset and conducted an initial inspection to examine variable types, missing values, and overall data structure. Date variables were converted into proper datetime formats to enable time-based analysis. Columns that were not informative for prediction, such as identifiers or redundant intermediate variables, were removed to reduce noise.

To improve data quality, missing values were handled carefully based on the nature of each variable. Numerical weather features with a small proportion of missing values were imputed using summary statistics, while records with excessive missing information were dropped to maintain data reliability. All weather-related numerical variables were converted to consistent units and data types to ensure compatibility across models.

Next, we created additional engineered features to better capture temporal and meteorological patterns. These included weekly and monthly aggregated variables, rolling averages of temperature and humidity, and lagged weather features to incorporate short-term historical effects. The target variable, *RainTomorrow*, was defined as a binary indicator representing whether rainfall occurs on the following day.

Finally, the dataset was split into training and testing subsets to support unbiased model evaluation. Feature scaling was applied where necessary to ensure fair comparison across models that are sensitive to variable magnitude.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine the statistical properties of the weather dataset, identify temporal and seasonal patterns, and understand the relationships between meteorological variables and rainfall occurrence. The findings from this section provide both empirical and conceptual motivation for feature engineering and model selection.

Distributional Characteristics and Class Balance

We first analyzed the distributions of key continuous variables, including temperature, wind speed, and precipitation. Temperature-related variables (TAVG, TMAX, TMIN) exhibit smooth, unimodal distributions, reflecting relatively stable daily thermal conditions. In contrast, precipitation-related variables (PRCP) are highly right-skewed, indicating that most days experience little or no rainfall, while a small subset of days accounts for heavy precipitation events.

An examination of the target variable reveals a noticeable class imbalance, with non-rainy days substantially outnumbering rainy days (approximately a 70/30 split). This imbalance suggests that naïve classifiers may bias toward predicting “no rain,” motivating careful model evaluation using recall-oriented metrics and the inclusion of temporal features that enhance sensitivity to rainfall events.

Correlation Structure and Multicollinearity

A correlation heatmap was generated to explore dependencies among weather variables. Strong multicollinearity is observed among temperature variables, with correlation coefficients exceeding 0.9 between TAVG, TMAX, and TMIN. This result implies that these variables encode overlapping thermal information, which may introduce redundancy in linear models but is less problematic for tree-based methods.

Wind-related variables (AWND, WSF2, WSF5) display moderate positive correlations with each other, suggesting shared information regarding atmospheric motion. Snow-related features (SNOW, SNWD) are correlated but occur infrequently, reflecting their seasonal nature. Precipitation-related variables show weaker contemporaneous correlations but gain explanatory power when temporal dependence is introduced.

Lagged features play an important role in capturing persistence in weather systems. PRCP_lag1 and RainYesterday exhibit positive associations with RainTomorrow, supporting the meteorological intuition that precipitation often occurs in clusters rather than isolated events. This persistence effect motivates the inclusion of lag variables in subsequent modeling.

Temporal Smoothing and Seasonality

To reduce daily noise and capture broader weather trends, rolling statistics were examined. Rolling averages of temperature smooth short-term fluctuations and provide a more stable representation of thermal conditions that influence atmospheric moisture capacity. Visual inspection confirms that smoothed temperature series better reflect sustained warming or cooling patterns than raw daily observations.

Seasonal patterns were further analyzed using monthly rainfall probability visualizations. The results reveal strong intra-year heterogeneity. For example, rainfall probability in 2024 peaks in April (approximately 0.45–0.50) and December (above 0.50), while October exhibits near-zero rainfall probability. Early 2025 data show elevated rainfall likelihood in late spring, reinforcing the role of seasonal cycles. These findings justify the inclusion of calendar-based features such as Month and DayOfYear.

Overall, the EDA highlights three key insights:

- (1) strong multicollinearity among temperature variables,
- (2) meaningful persistence captured by lagged precipitation features, and
- (3) pronounced seasonal variation in rainfall occurrence.

These insights directly inform feature engineering choices and model selection.

Model Training and Evaluation

Modeling Strategy and Setup

Rainfall prediction was formulated as a binary classification problem with *RainTomorrow* as the target variable. Multiple models were evaluated to balance interpretability and predictive performance, including Logistic Regression (L1 and L2 regularization), Decision Tree, and Random Forest classifiers. The dataset was split into training and testing sets using a fixed random seed. A consistent classification threshold of 0.4 was applied across all models to ensure comparability. Given the observed class imbalance, model performance was evaluated using multiple metrics, with recall selected as the primary metric, as failing to detect rainfall events is more costly than false alarms in practical forecasting contexts.

Baseline Models: Linear and Single-Tree Methods

Logistic Regression models with both L1 and L2 regularization achieved the highest overall accuracy (0.65) and strong AUC values (≈ 0.604). However, their recall remained limited (0.2468), indicating that linear decision boundaries struggle to capture the complex, non-linear interactions among weather variables. The Decision Tree model exhibited lower overall performance, with an accuracy of 0.5955, recall of 0.2338, and AUC of 0.5316. While the model can capture non-linear relationships, its sensitivity to training data suggests limited generalization ability when used alone.

Random Forest Performance and Cross-Validation

The Random Forest classifier demonstrated a more favorable balance between predictive accuracy and recall. The baseline Random Forest achieved an accuracy of 0.6318, recall of 0.3766, and AUC of 0.5989, outperforming all other models in terms of recall.

To improve robustness, hyperparameters were optimized using cross-validation. The CV-optimized Random Forest achieved the highest overall AUC (0.6077) while maintaining a recall of 0.3377. Although recall decreased slightly relative to the baseline Random Forest, the improvement in AUC indicates better discrimination across thresholds, reflecting enhanced ranking performance.

Feature Importance and Model Robustness

Feature importance analysis from the Random Forest model reveals that wind direction variables (WDF2, WDF5) are the most influential predictors, highlighting the importance of atmospheric circulation patterns. Seasonal variables (DayOfYear, Month) and temperature measures (TAVG, TMAX, TMIN) further contribute substantial predictive power, while precipitation persistence is captured by PRCP, PRCP_lag1, and RainYesterday.

A robustness check was conducted by training a simplified Random Forest using only the top five features. The simplified model achieved an accuracy of 0.6227 and AUC of 0.5905, compared to 0.6318 accuracy and 0.5989 AUC for the full model. Recall decreased modestly from 0.3766 to 0.3636,

indicating that most predictive power is concentrated in a small subset of key variables rather than dispersed across noise features.

Conclusion

Multiple classification models were trained and evaluated, including Logistic Regression, Decision Tree, and Random Forest. By comparing model performance using metrics such as accuracy, recall, and AUC-ROC, we found that ensemble-based models are more effective in capturing the non-linear relationships among weather variables. The Random Forest model demonstrated the most balanced and stable performance on unseen data and was selected as the final predictive model.

Key Findings:

Persistence is Key: "Yesterday's weather" is one of the best predictors of "tomorrow's weather," as evidenced by the high importance of lag variables in Figure 4.

Thermodynamics Matter: Temperature spread (TMAX/TMIN) and its rolling trends remain fundamental to precipitation logic.

Overall, the results indicate that historical weather conditions contain meaningful predictive signals for rainfall occurrence, and machine learning models can leverage these signals to produce reliable short-term rainfall predictions.

Future Work

While the current model demonstrates satisfactory performance; several directions may further enhance its applicability in real-world deployment. First, additional hyperparameter tuning could be conducted using systematic approaches such as GridSearch or RandomizedSearch to further optimize key Random Forest parameters, including tree depth and the number of estimators.

Second, more advanced temporal modeling techniques may be explored. Rather than relying on simple one-day lag features, recurrent architectures such as Long Short-Term Memory (LSTM) networks could better capture long-term sequential dependencies inherent in weather systems, potentially improving predictive accuracy over extended horizons.

Third, the integration of external data sources represents a promising avenue for improvement. Variables such as barometric pressure and humidity, which were not available in the current dataset, are known to be strong indicators of approaching storm fronts. Additional sources such as satellite imagery, atmospheric circulation indicators, or geographical features may provide complementary information that enhances model performance.

Fourth, future work could further address class imbalance, as rainfall events occur less frequently than non-rainy days. Techniques such as cost-sensitive learning, threshold adjustment, or resampling methods may help improve recall while maintaining overall model stability.

Finally, the current binary classification framework could be extended to multi-level rainfall intensity prediction, enabling more granular and actionable insights for downstream decision-making in practical forecasting applications.

Appendix 1 Variable Selection and Data Dictionary

The dataset used for this analysis includes daily climatic data from the New York City Central Park station. Below is a dictionary of the variables retained for analysis after initial filtering.

Variable	Description	DataType	Treatment
DATE	Date of observation	datetime64	Indexing
PRCP	Precipitation (tenths of mm)	float64	Target Derivation
SNOW	Snowfall (mm)	float64	FillNA(0)
SNWD	Snow depth (mm)	float64	FillNA(0)
TMAX	Maximum temperature (Fahrenheit)	float64	Interpolated
TMIN	Minimum temperature (Fahrenheit)	float64	Interpolated
AWND	Average daily wind speed	float64	Interpolated
WT01	Fog, ice fog, or freezing fog	float64	Binary Flag
WT02	Heavy fog or heaving freezing fog	float64	Binary Flag
WT03	Thunder	float64	Binary Flag
WT04	Ice pellets, sleet, snow pellets	float64	Binary Flag
WT05	Hail	float64	Binary Flag
WT06	Glaze or rime	float64	Binary Flag
WT08	Smoke or haze	float64	Binary Flag

Appendix 2 Data Cleaning Logic

The raw data contained missing values which were handled as follows to ensure model stability:

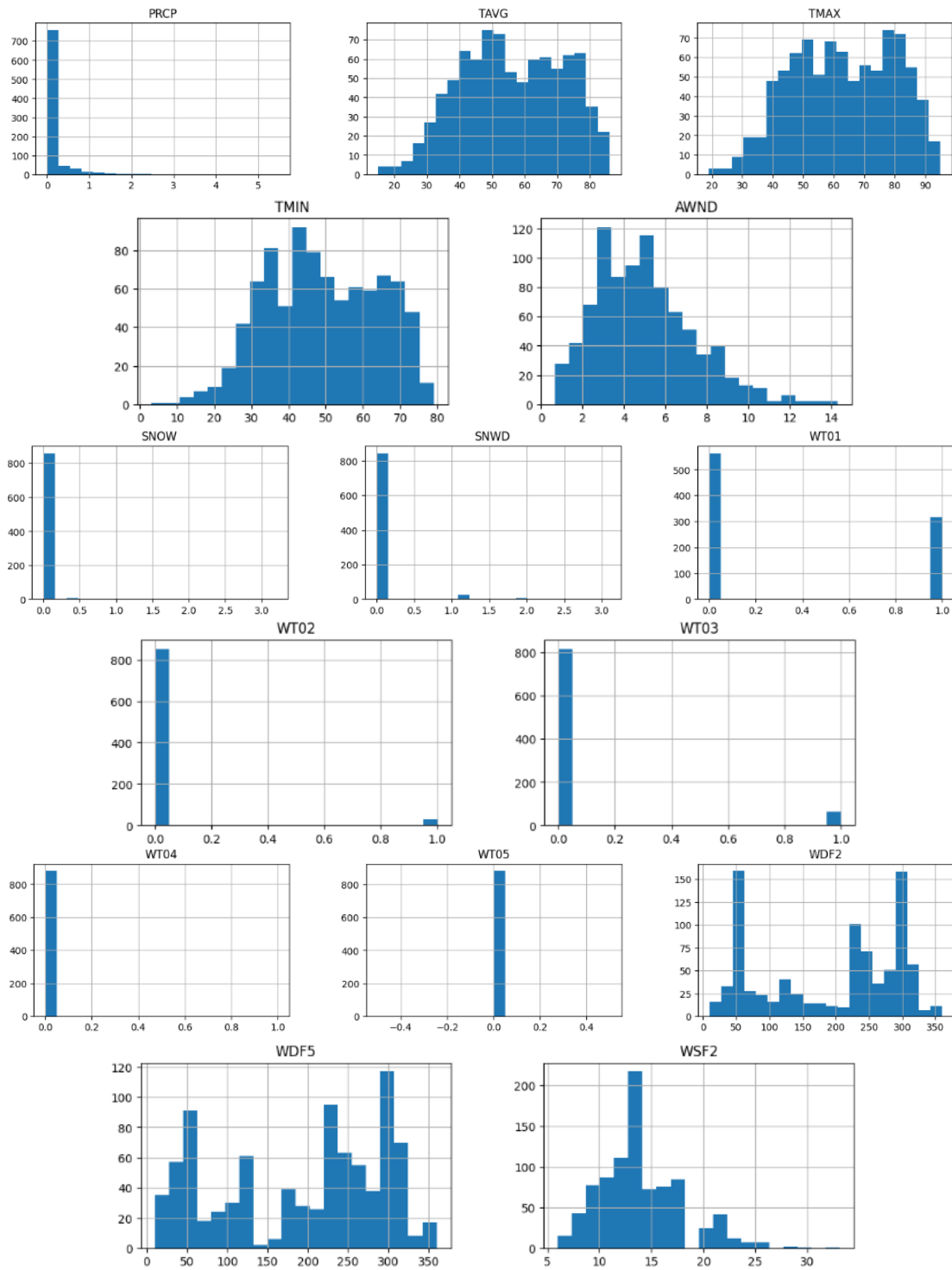
1. **Interpolation:** TMAX, TMIN, and AWND had minor gaps. These were filled using linear interpolation (`df.interpolate(method='linear')`) to preserve the natural weather trends between days.
2. **Zero-filling:** Missing values in precipitation (PRCP) and snow (SNOW, SNWD) columns were assumed to be days with zero occurrences and were filled with 0.
3. **Weather Types:** The WT** columns (Severe Weather Indicators) contained NaN where no event occurred. These were filled with 0 to represent the absence of the event.

Appendix 3 Model Specifications

The final Random Forest model was instantiated using the scikit-learn library with the following configurations. These parameters were chosen to balance performance and computational efficiency.

```
RandomForestClassifier(
    n_estimators=100,    # Number
                        of trees in the forest
    criterion='gini',    # Splitting
                        criterion
    max_depth=None,      # Nodes are expanded until all
                        leaves are pure
    min_samples_split=2, # Minimum
                        number of samples required to split
    min_samples_leaf=1,  # Minimum number of samples
                        required at a leaf node
    bootstrap=True,      #
                        Bootstrap samples are used when building trees
    random_state=42,     # Seed for reproducibility
    class_weight=None     # Weights associated with classes (handled via SMOTE)
)
```

Appendix 4



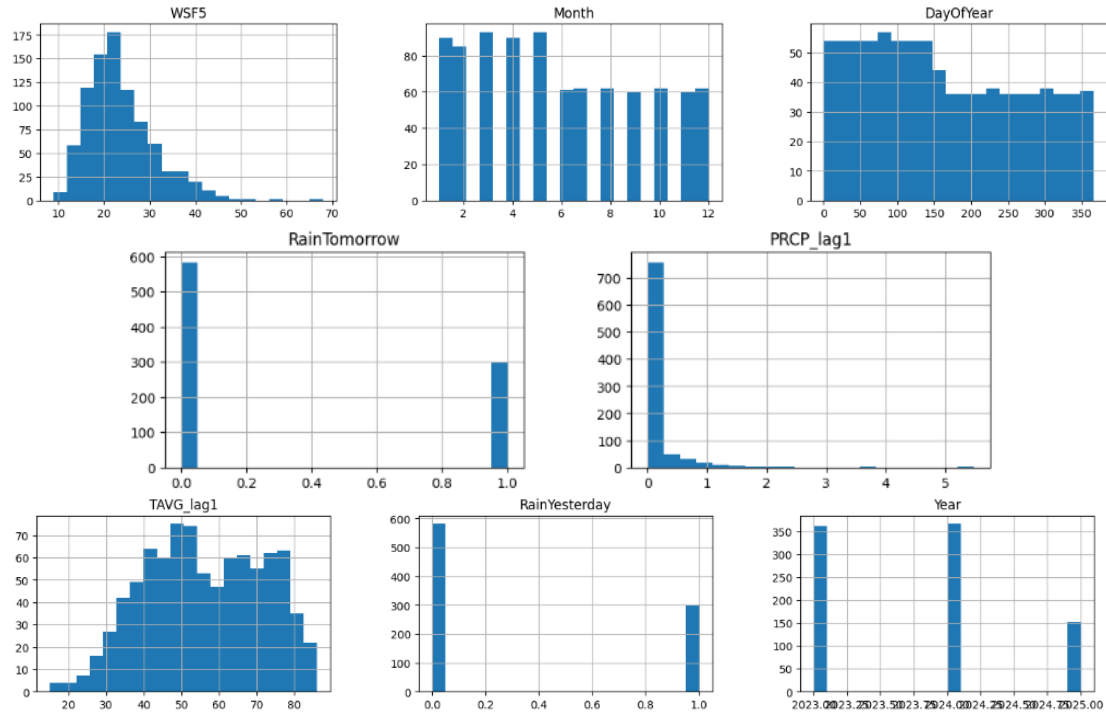


Figure 1: Histograms of numerical features in the dataset

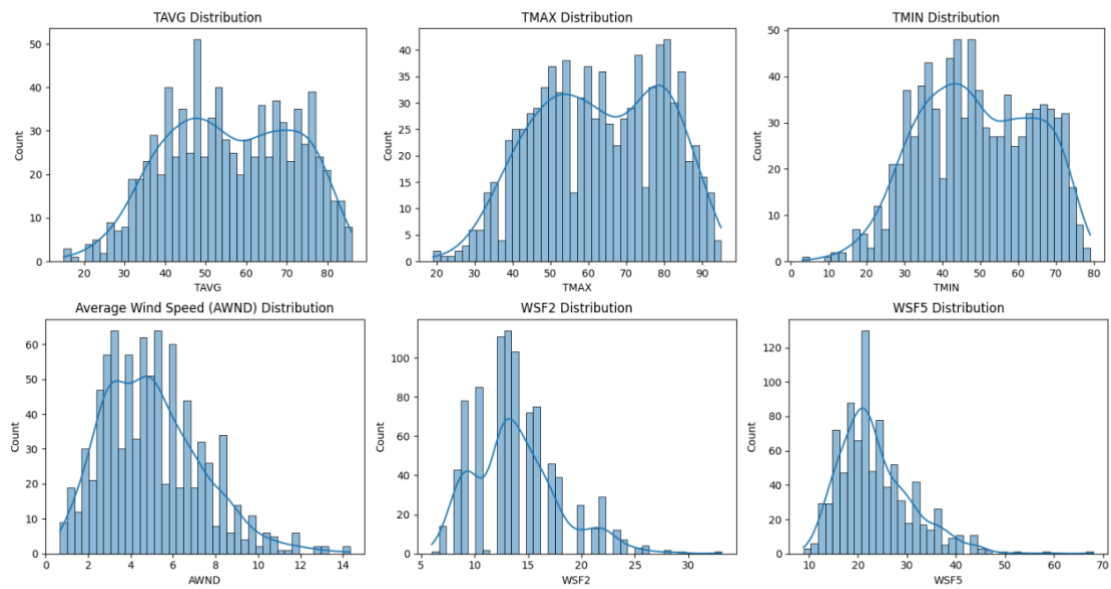


Figure 2: Distributions of selected temperature and wind-related variables

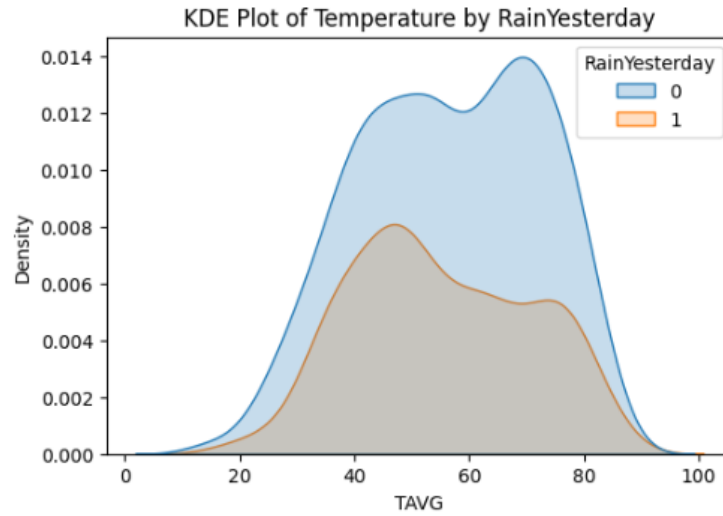


Figure 3: Kernel density plot of average temperature (TAVG) grouped by RainYesterday

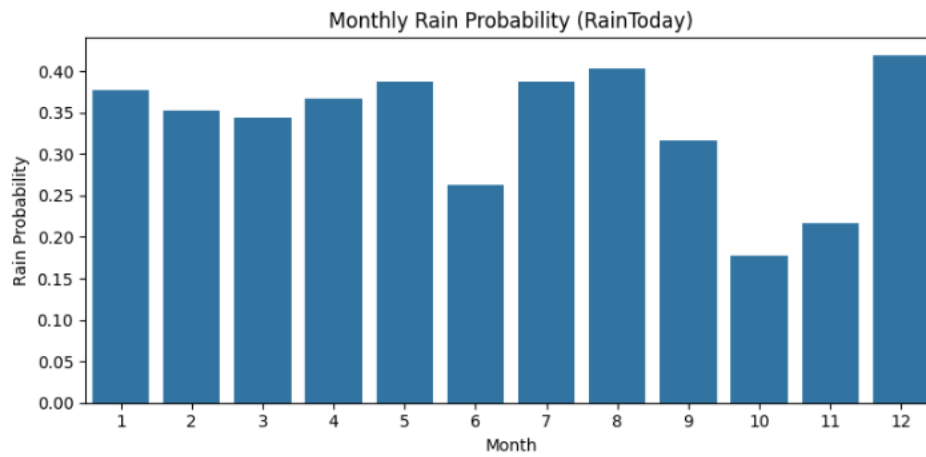


Figure 4: Monthly rainfall probability calculated from historical data

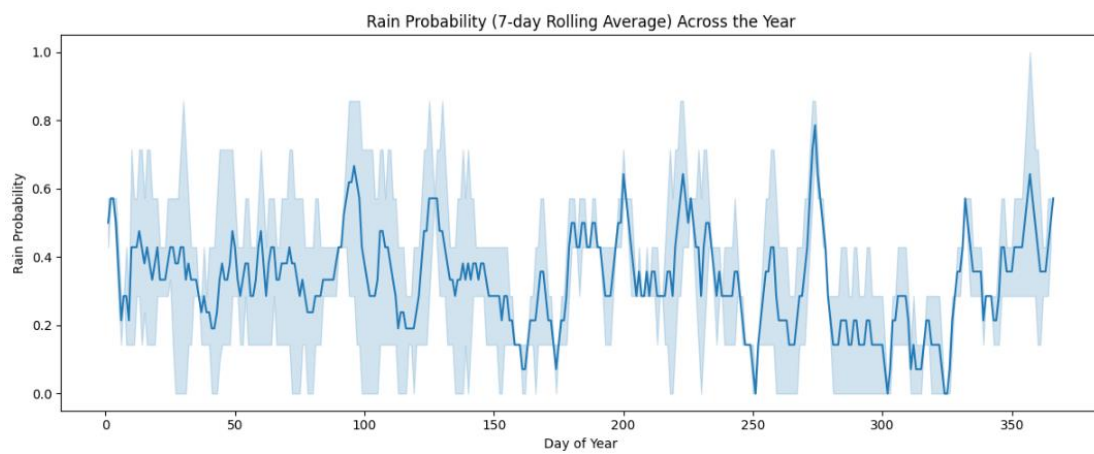


Figure 5: Rainfall probability by day of year using a 7-day rolling average

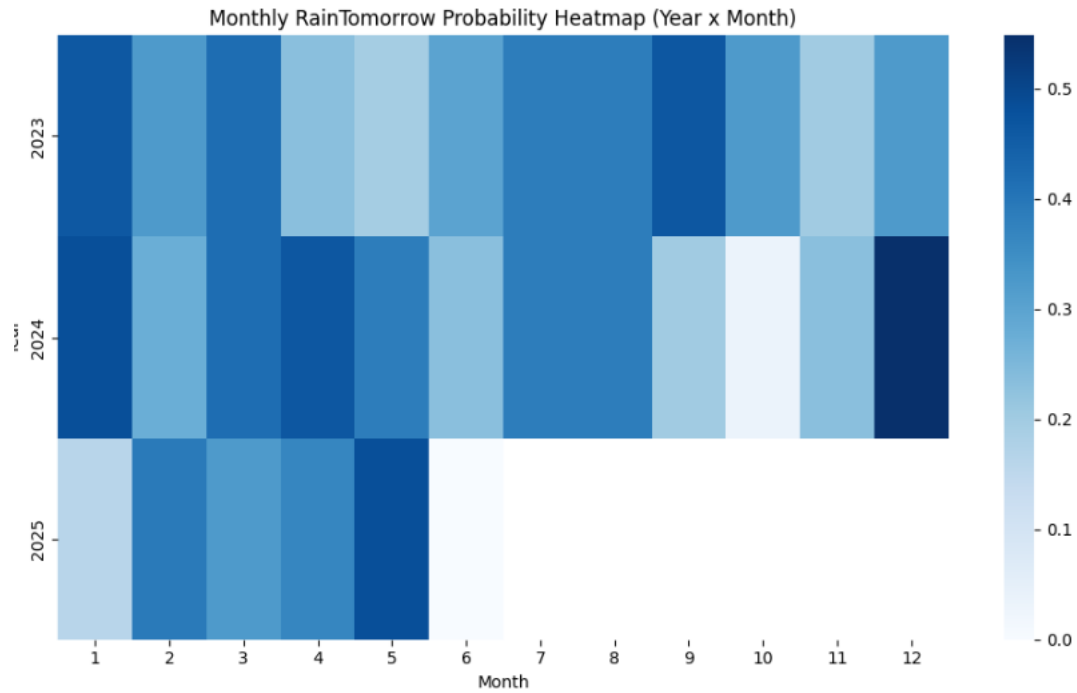


Figure 6: Heatmap of monthly RainTomorrow probability by year

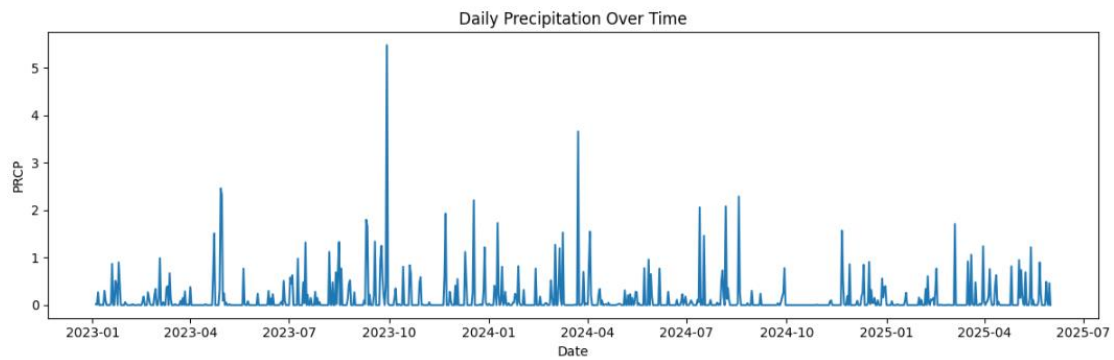


Figure 7: Time series of daily precipitation (PRCP)

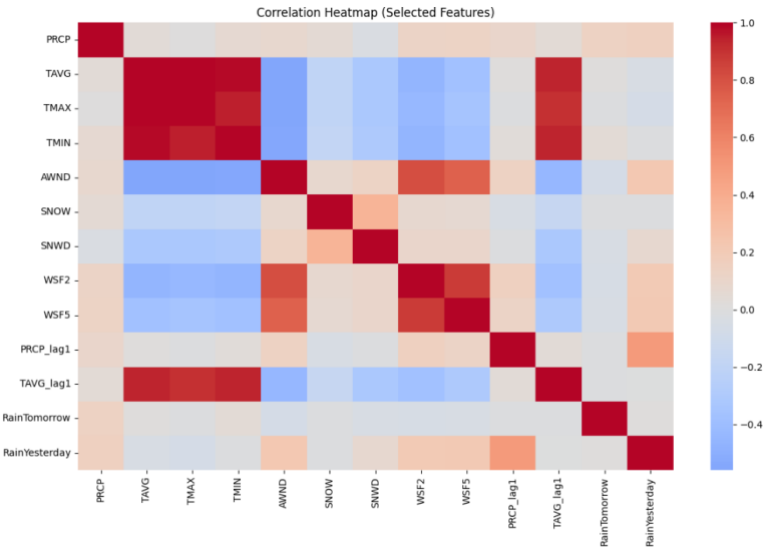


Figure 8: Correlation heatmap of selected weather and lagged features.

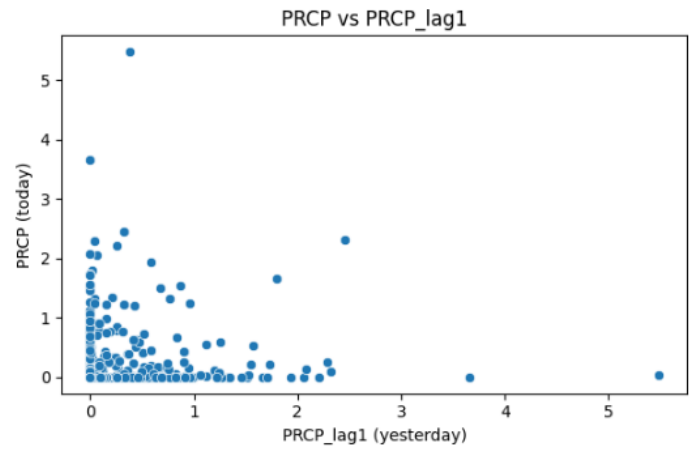


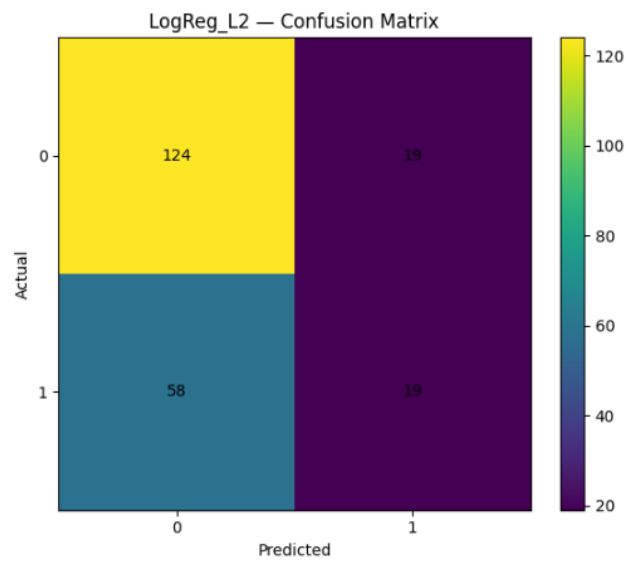
Figure 9: Scatter plot of PRCP versus lagged precipitation (PRCP_lag1).

Metric preference = RECALL | threshold = 0.4

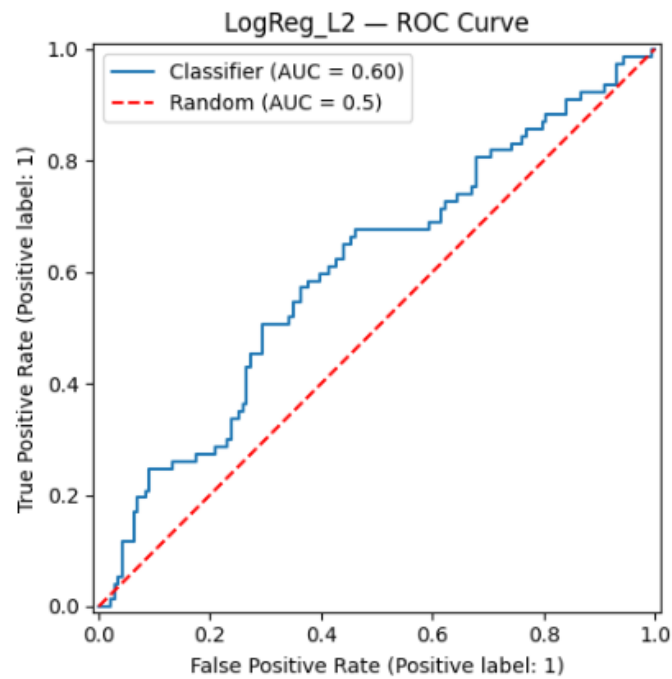
	accuracy	precision	recall	f1	auc	threshold	primary_metric
model							
RandomForest	0.6318	0.4677	0.3766	0.4173	0.5992	0.4	0.3766
LogReg_L2	0.6500	0.5000	0.2468	0.3304	0.6038	0.4	0.2468
LogReg_L1	0.6500	0.5000	0.2468	0.3304	0.6035	0.4	0.2468
DecisionTree	0.5955	0.3750	0.2338	0.2880	0.5316	0.4	0.2338

Figure 10: Model performance table reporting accuracy, precision, recall, F1 score, and AUC at a fixed threshold of 0.4.

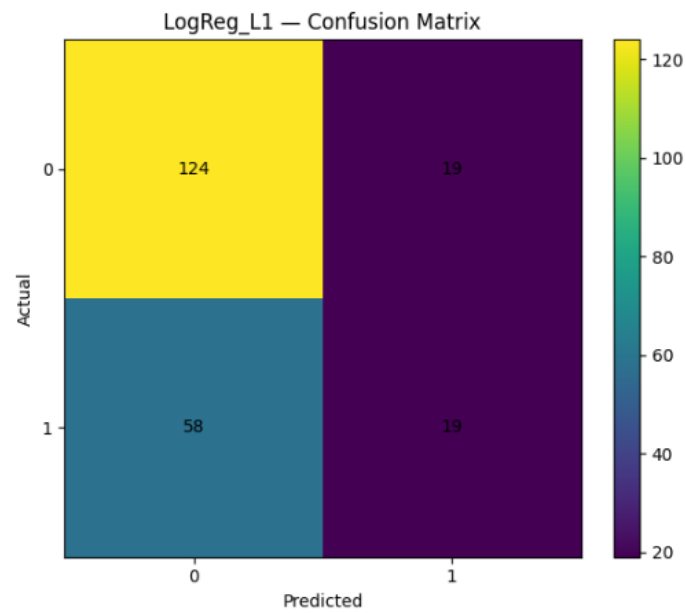
=== LogReg_L2 ===



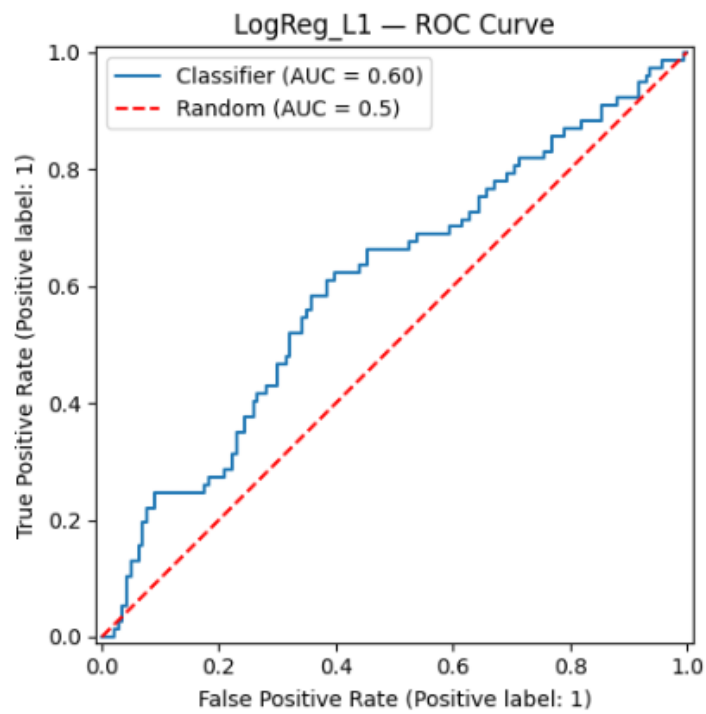
<Figure size 600x500 with 0 Axes>



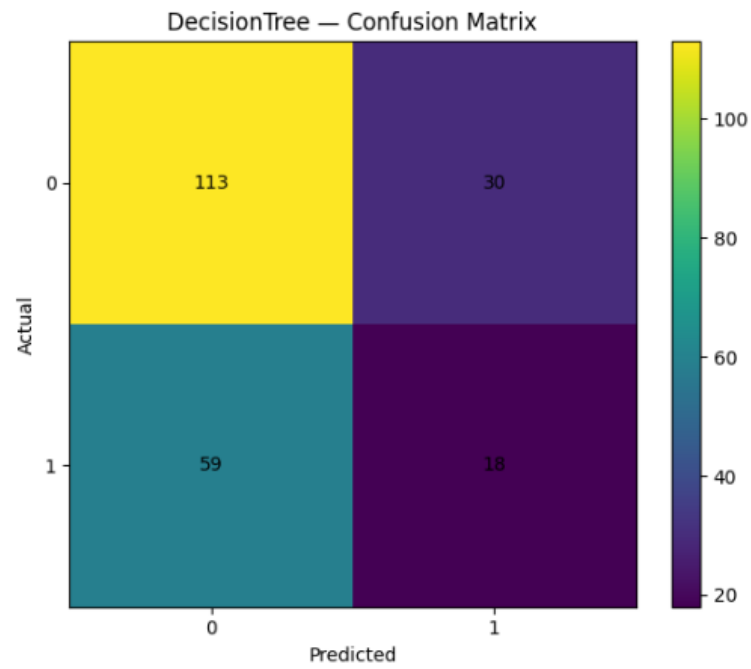
=== LogReg_L1 ===



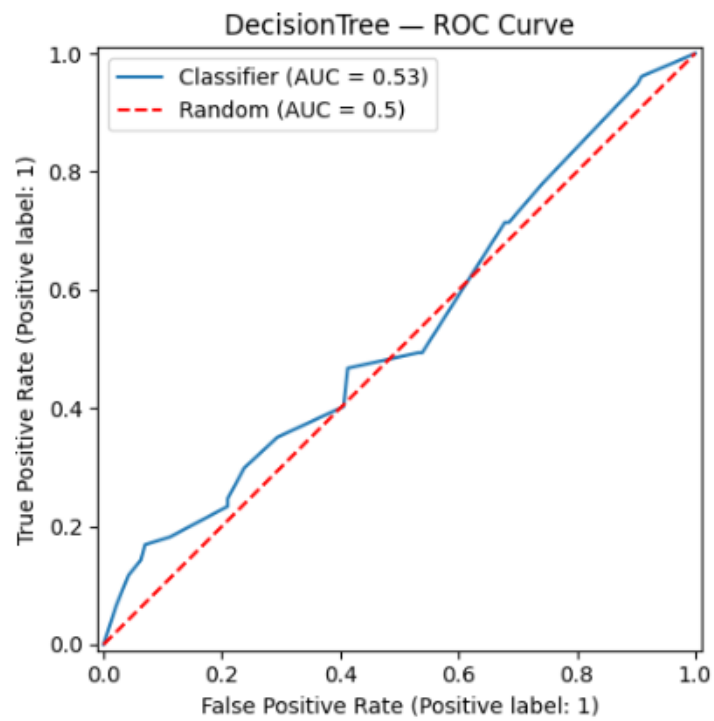
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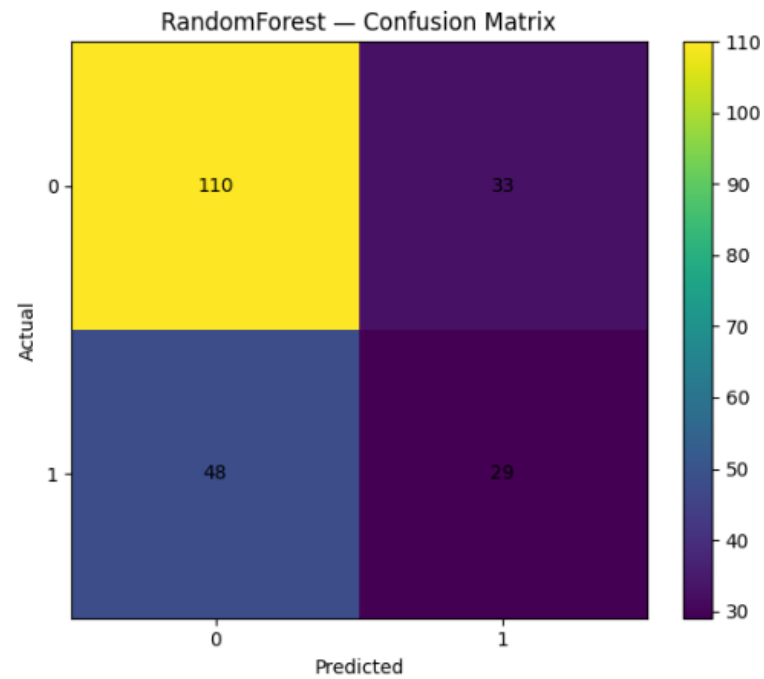
```
=== DecisionTree ===
```



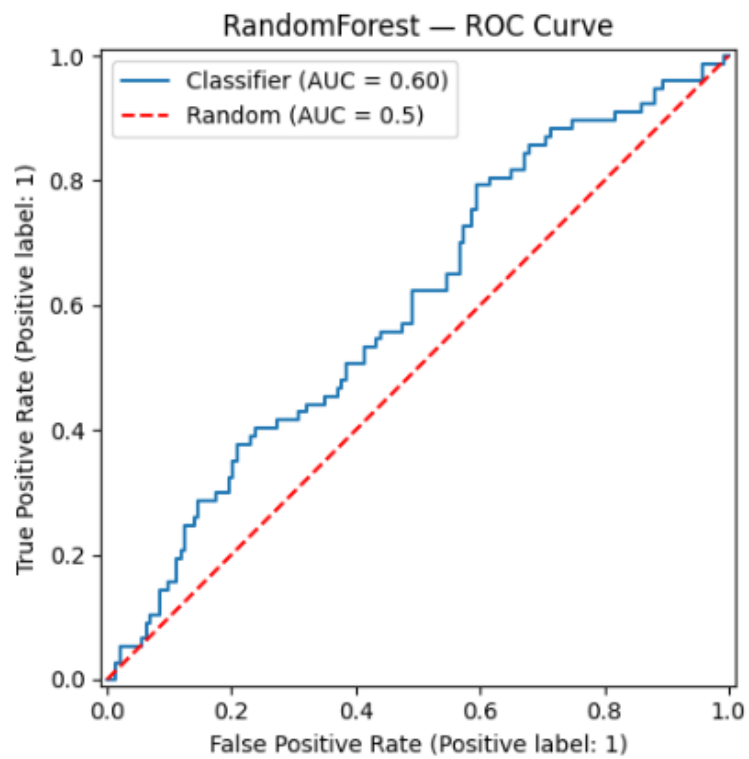
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=== RandomForest ===

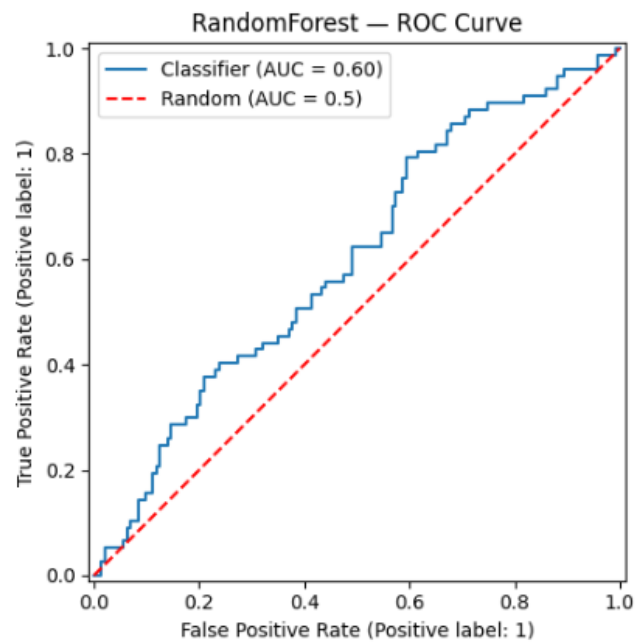
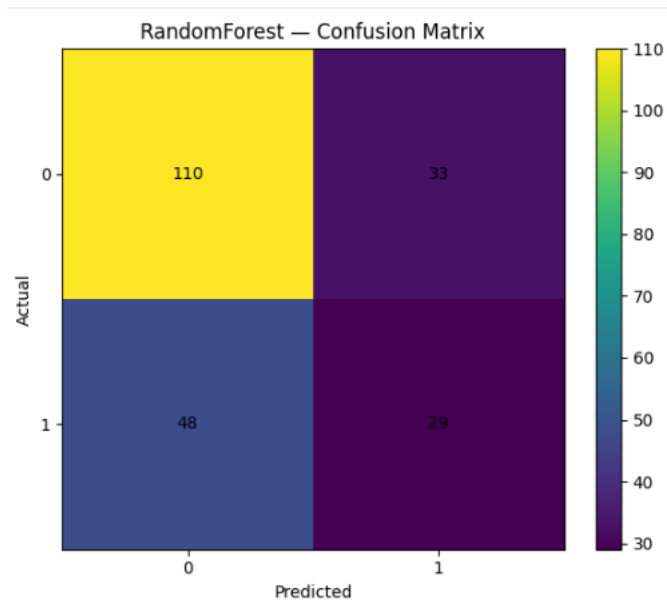


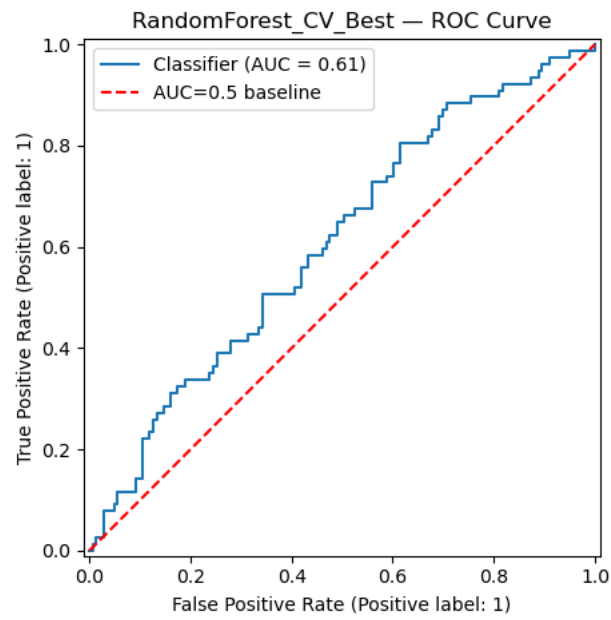
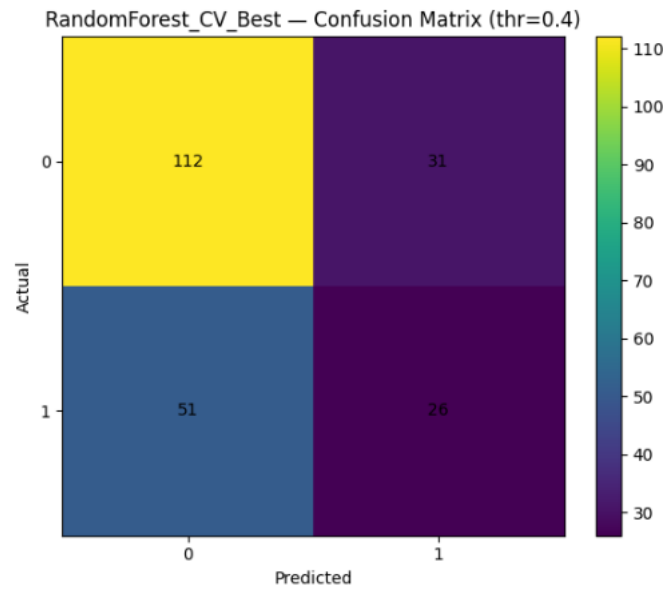
<Figure size 600x500 with 0 Axes>



	accuracy	precision	recall	f1	auc	threshold	primary_metric
model							
RandomForest	0.6318	0.4677	0.3766	0.4173	0.5992	0.4	0.3766
LogReg_L2	0.6500	0.5000	0.2468	0.3304	0.6038	0.4	0.2468
LogReg_L1	0.6500	0.5000	0.2468	0.3304	0.6035	0.4	0.2468
DecisionTree	0.5955	0.3750	0.2338	0.2880	0.5316	0.4	0.2338

Figure: Model performance comparison table across different classification models evaluated at a threshold of 0.4.





	accuracy	precision	recall	f1	auc	threshold	primary_metric
RandomForest	0.6318	0.4677	0.3766	0.4173	0.5992	0.4	0.3766
RandomForest_CV_Best	0.6273	0.4561	0.3377	0.3881	0.6065	0.4	0.3377
LogReg_L2	0.6500	0.5000	0.2468	0.3304	0.6038	0.4	0.2468
LogReg_L1	0.6500	0.5000	0.2468	0.3304	0.6035	0.4	0.2468
DecisionTree	0.5955	0.3750	0.2338	0.2880	0.5316	0.4	0.2338

Figure: Model performance table including the cross-validated Random Forest model evaluated at a threshold of 0.4.

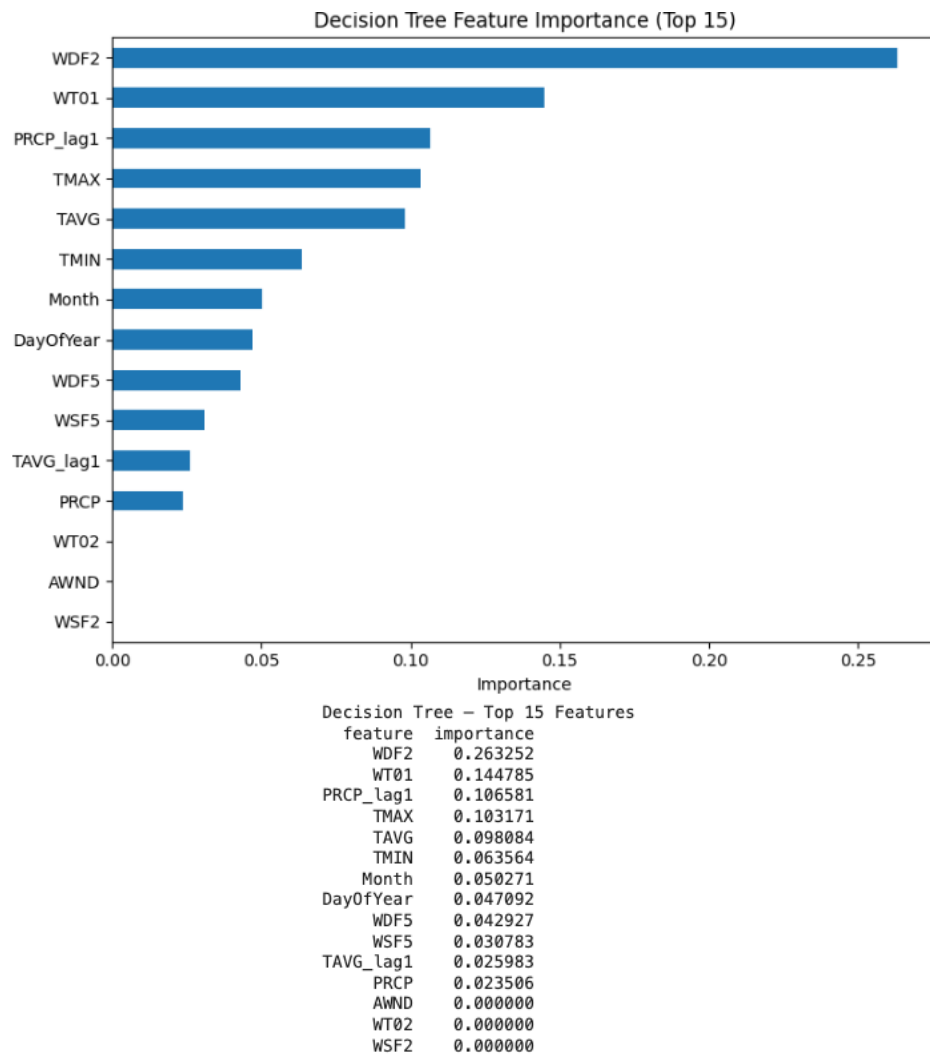


Figure: Feature importance plot for the Decision Tree model showing the top 15 predictors

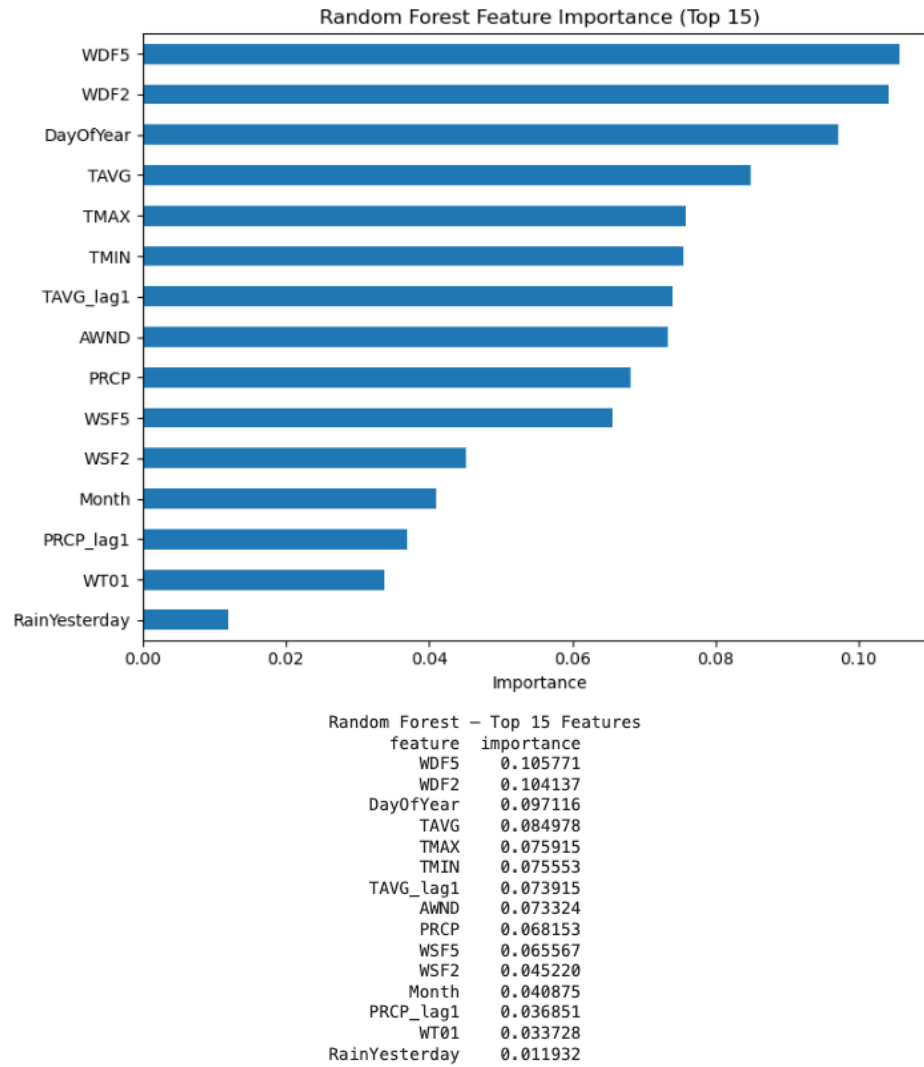


Figure: Feature importance plot for the Random Forest model showing the top 15 predictors.

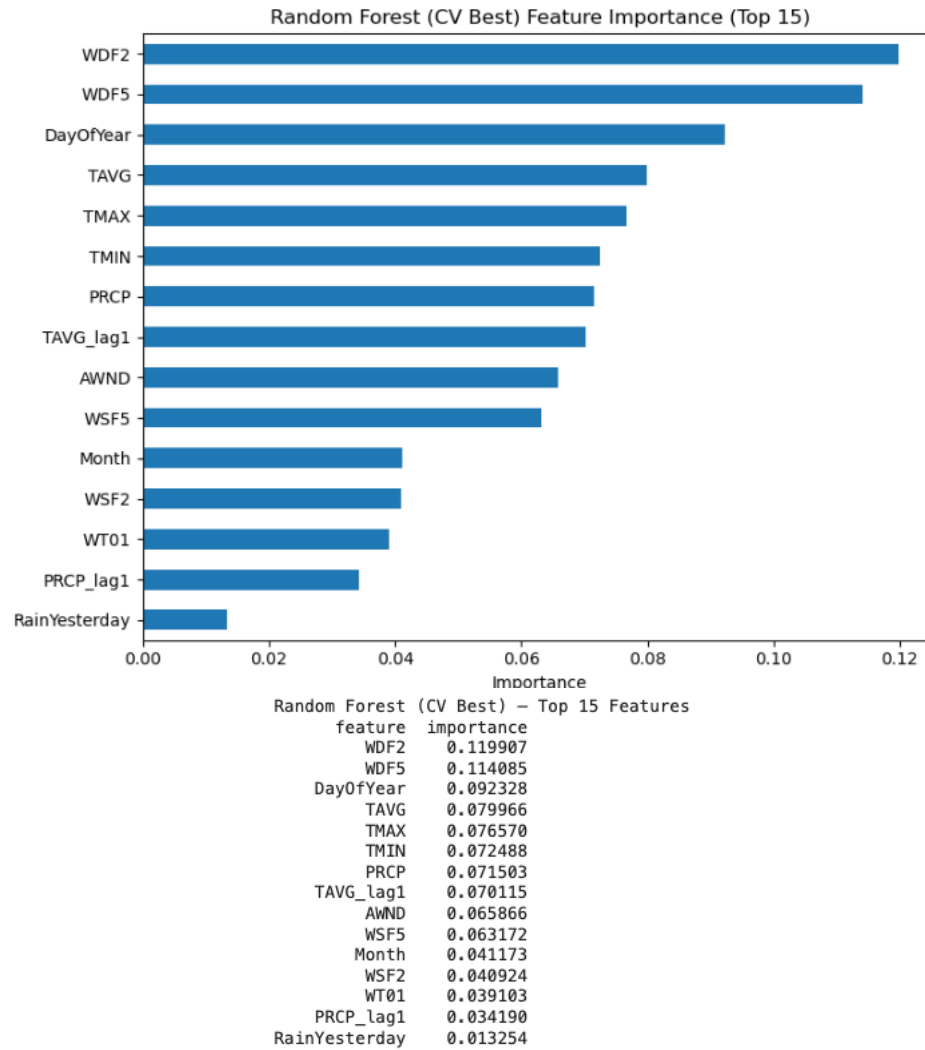
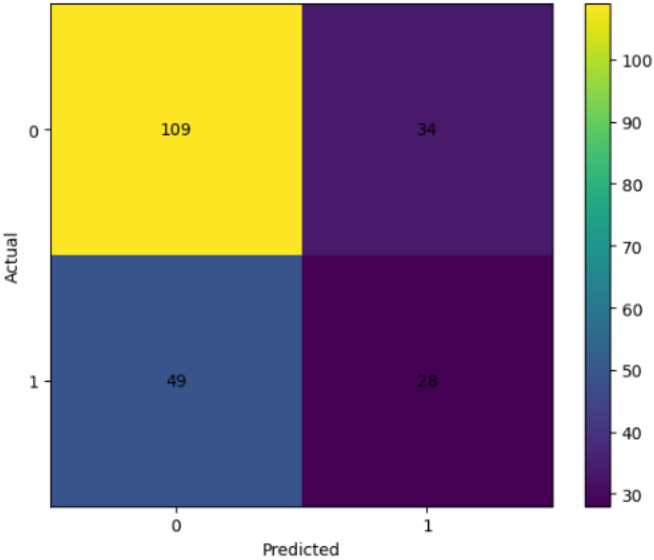


Figure: Feature importance plot for the cross-validated Random Forest model showing the top 15 predictors.

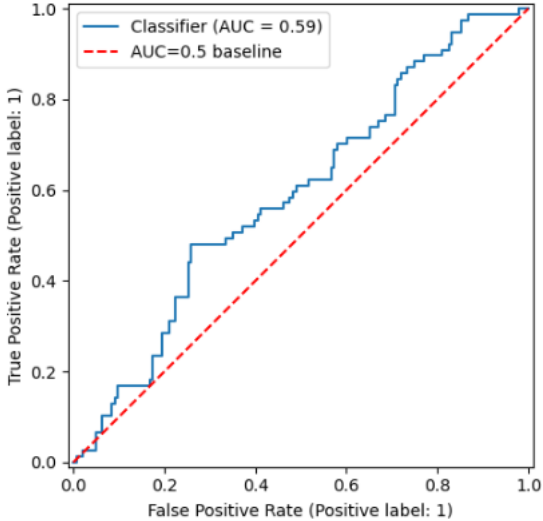
Top 5 Features used for robustness check:		Simplified RF (Top 5 features) – Test Performance	
- WDF2		accuracy:	0.6227
- WDF5		precision:	0.4516
- DayOfYear		recall:	0.3636
- TAVG		f1:	0.4029
- TMAX		auc:	0.5905
=== Robustness Check: Full RF vs Simplified RF ===			
		Full RF (All Features)	Simplified RF (Top 5)
accuracy		0.6318	0.6227
precision		0.4677	0.4516
recall		0.3766	0.3636
f1		0.4173	0.4029
auc		0.5989	0.5905

Figure: Robustness check results for a simplified Random Forest model using the top five features, reporting test-set performance metrics.

Simplified RF (Top 5 Features) — Confusion Matrix (thr=0.4)



Simplified RF (Top 5 Features) — ROC Curve



<Figure size 700x600 with 0 Axes>